Global Gas Flaring Analysis

Flare Volume, Flare Intensity, and Oil Production Trends, 2012–2025

Personal Analytical Project
Prepared using Microsoft Excel (Power Query, PivotTables, PivotCharts)

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1. Executive Summary

This report presents an analysis of global gas flaring activity between 2012 and 2025, covering 107 countries. The analysis draws on three linked datasets — flare volume, flare intensity, and oil production — merged, cleaned, and validated into a single master dataset for the purpose of identifying global trends, country-level contributors, and data-quality irregularities worth disclosing to any reader of the findings.

Global flare volume rose from 143.68 billion cubic metres (bcm) in 2012 to a pre-pandemic peak of 173.62 bcm in 2019, fell sharply to 143.38 bcm by 2022 amid the global oil-demand contraction, and has since recovered to 166.55 bcm in 2025 — a net increase of approximately 15.9% over the full period. Cumulative flare volume across all 14 years totalled 2,166.44 bcm, against a cumulative oil production figure of 1,122,365.44 (thousand barrels/day, summed across years).

A small group of countries account for the large majority of global flaring. Iraq, Russia, Iran, and Venezuela alone contribute approximately 60.7% of total flare volume across the period, with Iraq the single largest contributor at 18.17%.

2. Introduction

2.1 Purpose

This workbook analyzes gas-flaring trends — flare volume, flare intensity, and oil production — across countries from 2012 to 2025, with the aim of identifying global and country-level patterns, and building a dashboard suitable for ongoing monitoring.

2.2 Scope and Data Coverage

The dataset spans 2012–2025 (14 years) and covers 107 countries. Of these, 104 countries recorded gas flaring activity at least once during the period, while a smaller subset recorded no flaring throughout. Separately, 91 countries recorded oil production activity, meaning 16 countries in the dataset had no recorded oil production over the full period.

2.3 Key Variables

Variable	Unit	Description
Flare Volume	BCM (Billion Cubic Metres)	Volume of associated gas flared per country per year
Flare Intensity	m³/bbl (Cubic Metres per Barrel)	Gas flared per barrel of oil produced — an efficiency measure
Oil Production	kbbl/day (Thousand Barrels per Day)	Crude oil production rate per country per year

3. Methodology

3.1 Data Preparation

Three original source files — Flare Volume, Flaring Intensity, and Oil Production — were imported and cleaned independently before being merged on Country and Year into a single Master Dataset. Cleaning was performed using Power Query.

3.2 Merge Integrity

An early data-quality issue was identified during merging: rows belonging to countries present in the Oil Production table but absent from the Flare Intensity/Volume tables (Cyprus, Czechia, and Suriname) initially lost their Country and Year identifiers after a Full Outer join, because these fields were expanded from only one side of the merge. This was corrected by expanding the joining table's own Country and Year fields and coalescing them with the primary fields, restoring full row identity across the merged dataset.

3.3 Treatment of Zero Values

Zero values in the dataset were deliberately retained rather than removed or imputed, based on the principle that a zero value can represent one of two materially different situations:

- Genuine zero flaring — a country with active, non-trivial oil production but no recorded flaring, indicating effective associated-gas capture.
- Flaring without matching oil production — a small number of countries (e.g. Ethiopia, Cambodia) show occasional non-zero flare readings despite zero recorded oil production, consistent with satellite-detected flaring unrelated to the oil production series captured in this dataset.

Rather than silently editing or removing ambiguous rows, three flag columns — WITHOUT FLARE INTENSITY, WITHOUT FLARE VOLUME, and WITHOUT OIL PRODUCTION — were added to the master dataset to make every zero-value case identifiable and auditable. Missing values were treated as part of the zero-flare country group, per the workbook's documented convention.

3.4 Analytical Tools

Analysis was performed using native Excel PivotTables and PivotCharts, with a dedicated Timeline control (built from a converted date field) enabling year-range filtering across the dashboard, and a Country slicer enabling country-level drill-down.

4. Key Findings

4.1 Global Trend Summary

Metric	Value
Total Flare Volume (2012–2025, summed)	2,166.44 bcm
Total Oil Production (2012–2025, summed)	1,122,365.44 (kbbl/day, summed across years)
Average Flare Intensity (2012–2025)	5.29 m³/bbl
2012 Flare Volume	143.68 bcm
2025 Flare Volume	166.55 bcm
Overall % Change, 2012 → 2025	+15.9%
Peak Year	2019 (173.62 bcm)
Countries Tracked	107
Countries with Recorded Flaring	104
Oil-Producing Countries	91

4.2 Trend Narrative

Global flare volume followed a broadly upward trajectory from 2012 to a peak in 2019, before falling sharply through 2020–2022 — a period coinciding with the global contraction in oil demand and production during the COVID-19 pandemic. Flaring has since recovered steadily from its 2022 low (143.38 bcm) back to 166.55 bcm in 2025, though this remains slightly below the 2019 peak. This indicates that, while global flaring is trending upward again, it has not yet fully returned to its pre-pandemic high.

4.3 Leading Contributors

A small number of countries account for the large majority of global flare volume across the period:

Rank	Country	Share of Total Flare Volume (2012–2025)
1	Iraq	18.17%
2	Russia	16.62%
3	Iran	14.60%
4	Venezuela	11.29%
5	Algeria	7.31%
6	Nigeria	5.15%
7	Mexico	4.35%
8	Libya	4.13%

Rank	Country	Share of Total Flare Volume (2012–2025)
9	United States	3.86%
10	Angola	2.21%

Together, these ten countries account for approximately 87.7% of total flare volume recorded across the entire dataset, while the remaining 94 flaring countries collectively contribute the balance.

5. Dashboard Overview

The accompanying workbook includes an interactive dashboard ("My DashBoard" sheet) built around the following components:

- KPI summary cards — Total Oil Production, Total Flare Volume, Countries Tracked, Average Flare Intensity, and headline percentage-change indicators.
- Global trend charts — separate line/area charts for Flare Volume, Flare Intensity, and Oil Production, each spanning the full 2012–2025 range.
- Top-N country charts — Annual Flare Volume, Annual Oil Production, and Annual Flare Intensity, each ranked independently by their own metric.
- A Percentage Contribution chart — showing the top 20 flaring countries' share of total flare volume.
- A geographic map — visualizing the top 10 flaring countries by location.
- An interactive Timeline and Country slicer, enabling year-range and country-level filtering across connected visuals.

A documented limitation: because the three "Annual" charts (Flare Volume, Oil Production, Flare Intensity) are ranked independently, the countries appearing in each chart's Top N may differ from one another. Comparing a specific country across all three metrics currently requires manually selecting that country via the slicer, rather than being automatically synchronized.

6. Limitations and Notes

- The dataset does not distinguish routine from non-routine flaring, meaning the flare-volume trend should be read as an indicator of overall flaring activity rather than a direct measure of progress toward the World Bank's Zero Routine Flaring by 2030 initiative.
- Flare volume, flare intensity, and oil production coverage is not fully consistent across all 107 countries — a small number of countries appear in only one or two of the three original source tables.
- Zero values are retained as-is per the methodology in Section 3.3; readers comparing raw totals should be aware that some zero entries reflect absence of matching data rather than confirmed zero activity.
- A unified relative-change (indexed) view comparing Flare Volume, Flare Intensity, and Oil Production on a common 2012-baseline scale was scoped but not included in the current version of the dashboard.

7. Conclusion

Global gas flaring in 2025 remains close to, but below, its 2019 peak, following a pandemic-driven contraction and a subsequent multi-year recovery. Flaring activity remains heavily concentrated: ten countries account for roughly 88% of the global total, with Iraq, Russia, Iran, and Venezuela consistently the largest contributors across the period. The underlying dataset, once cleaned and validated, supports reliable year-over-year and country-level comparison, with data-quality caveats transparently documented throughout this report and the accompanying workbook.